The documentation and process conversion measures necessary to comply with this document shall be completed by 23 August 2022.

INCH-POUND
MIL-PRF-19500/717A
23 May 2022
SUPERSEDING
MIL-PRF-19500/717
9 May 2005

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, NON-HERMETIC, EPOXY, SURFACE MOUNT, SILICON, BIDIRECTIONAL TRANSIENT VOLTAGE SUPPRESSOR, TYPES 1N6036AUG THROUGH 1N6072AUG, 1N6036AUJ THROUGH 1N6072AUJ, JANP, JANPTX, AND JANPTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for non-hermetic, epoxy, surface mount 1,500 watt, bi-directional silicon transient voltage suppressor diodes. Three levels of product assurance (JANP, JANPTX and JANPTXV) are provided for each device as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (surface mount).

1.3 Maximum ratings. Maximum ratings are as shown in maximum test ratings (see 3.7) and as follows:

P_{PP} = 1500 W (see figures 2 and 3) at t_p = 1.0 ms.

$P_{M(AV)}$ = 1.0 W (derate at 8 mW/°C above T_A = +25°C).

I_{FSM} = 200 A (pk) at t_p = 8.3 ms (T_A = +25°C).

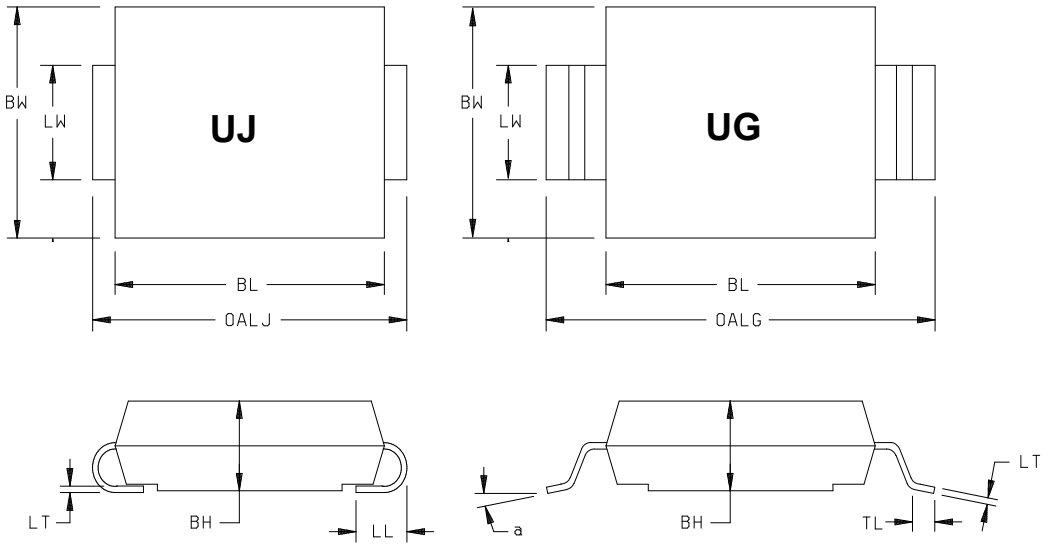
* -55°C ≤ T_J ≤ +150°C, -55°C ≤ T_{STG} ≤ +150°C (ambient).

1.4 Primary electrical characteristics at T_A = +25°C. Primary ratings are as shown in maximum test ratings (see 3.8).

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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Symbol	Inches	Inches	mm	mm	Notes
	Min	Max	Min	Max	
BH	.075	.103	1.91	2.62	
BL	.260	.280	6.60	7.11	
BW	.220	.245	5.59	6.22	
LL	.030	.060	0.76	1.52	
LT	.006	.012	0.15	0.30	
LW	.108	.124	2.74	3.15	
OALG	.380	.400	9.65	10.16	
OALJ	.305	.320	7.75	8.13	
TL	.025	.040	.635	1.02	
a		10°		10°	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
* 2. UG indicates Gull-Wing lead configuration; UJ indicates a J-Lead configuration.
* 3. In accordance with ASME Y14.5, measurements are equivalent to Φx symbology.

FIGURE 1. Physical dimensions.

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* 1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

* 1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for non-hermetic devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANP", "JANPTX" and "JANPTXV".

* 1.5.2 Device type. The designation system for the device types of diodes covered by this specification sheet are as follows.

* 1.5.2.1 First Number and first letter symbol. The diodes of this specification sheet use the letter symbols "1N".

* 1.5.2.2 Second Number symbols. The second number symbols covered by this specification are as follows: "6036" through "6072".

* 1.5.2.3 First suffix symbols. Following the second number symbols, the suffix letter "A" is used on devices that have a 5 percent voltage tolerance.

* 1.5.2.4 Second suffix symbols. Following the first suffix symbols, the suffix letter "UG" is used on devices that use a gullwing package and "UJ" is used on devices that use a J wing package.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, and 4, of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, and 4, of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://quicksearch.dla.mil.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

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* 3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$I_{(BR)}$	Reverse breakdown current at the specified condition.
P_{PP}	Peak pulse power.
$P_{M(AV)}$	Rated average power dissipation. This is comparable to the dc power rating of the device.
$V_{C(MAX)}$	Maximum clamping voltage. The maximum peak voltage appearing across the device when subjected to the peak pulse current I_P.
$\alpha_{V(BR)}$	Temperature coefficient of $V_{(BR)}$.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

* 3.4.1 Metallurgically bonded construction. Devices shall be metallurgically bonded, non-cavity, construction in accordance with the requirements of category II (see MIL-PRF-19500), and herein.

3.4.2 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

* 3.5 Marking. Devices shall be marked in accordance with MIL-PRF-19500. For surface mount devices, the minimum marking shall be PIN (1N and UG or UJ may be omitted), manufacturer's name, trademark, or identification or manufacturer's designating symbol, lot identification code, and code for plants. All marking which is omitted from the body of the device shall appear on the label of the initial container. The part number may be reduced to JPxxxx, JPXxxxx, or JPVxxxx.

3.6 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in 1.3, 1.4, and tables I, II, and IV.

* 3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in tables I and II herein.

3.8 Maximum test ratings and primary electrical characteristics. Maximum ratings shall be in accordance with columns 5 through 8 of table IV. Primary electrical characteristics are as shown in columns 2 and 4 of table IV.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

* 4.1.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-19500 and as specified herein except lot accumulation shall be 6 months in lieu of 6 weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or requalification only. In case qualification was awarded to a prior revision of the associated specification that did not request the performance of table II tests, the tests specified in table II herein shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANPTX and JANPTXV levels only). Screening shall be in accordance with table J-II of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement
	JANPTX and JANPTXV level
3a	$T_{(high)}$ = +125°C
7, 9, 10, 11	Not applicable
12	See 4.3.1
13	Interim electrical, delta, and table I, subgroups 2 and 3, electrical parameters not applicable for this screen (performed in screen 12).
14	Not required
15a, 15b	Not applicable

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* 4.3.1 Power burn-in (HTRB) and steady-state operation life test conditions. The test conditions and order of events shall be as follows:

a. Pulse in accordance with 4.5.2.a, in polarity A 20 times (screening and group B) and 100 times (group C and E) at T_A = +25°C.

b. Pulse in accordance with 4.5.2.a, in polarity B 20 times (screening and group B) and 100 times (group C and E) at T_A = +25°C.

c. Read I_D and $V_{(BR)}$ in polarities A and B at T_A = +25°C, remove defective devices and record the number of failures.

d. Apply the working peak reverse voltage (V_{WM}) (column 4 of table IV) at polarity A at T_A = +125°C as follows:

(1) For 96 hours (JANPTX and JANPTXV) for the screening test.

(2) For 340 hours (JANPTX and JANPTXV) for group B steady-state operation life test.

(3) For 1000 hours for group C steady-state operation life test.

e. Read I_D in polarity A at T_A = +25°C. Devices with ΔI_D >50 percent (100 percent for steady-state operation life) of the initial reading or 1 μA dc, whichever is greater shall be considered defective. Remove defective devices and record the number of failures (see note).

f. Apply the working peak reverse voltage (V_{WM}) (column 4 of table IV) at polarity B at T_A = +125°C as follows:

(1) For 96 hours (JANPTX and JANPTXV) for the screening test.

(2) For 340 hours (JANPTX and JANPTXV) for group B steady-state operation life test.

(3) For 1000 hours for group C steady-state operation life test.

g. Read I_D in polarity B at T_A = +25°C. Devices with ΔI_D >50 percent (100 percent for steady-state operation life) of the initial reading or 1 μA dc, whichever is greater shall be considered defective. Remove defective devices and record the number of failures (see note).

h. Read V_{BR} in polarities A and B at T_A = +25°C. Devices with ΔV_{BR} >2 percent (± 5 percent for steady-state operation life) of the initial reading shall be considered defective. Remove defective devices and record the number of failures (see note).

i. Read I_D in polarity A at T_A = +25°C, remove defective devices and record the number of failures.

NOTE: For the purpose of this test, the direction in which the device is first pulsed shall be considered polarity A and the reverse direction polarity B.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. End-point electrical measurements shall be in accordance with table I, subgroup 2 herein.

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* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table J-IVB (JANP, JANPTX and JANPTXV) of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, table J-IVB (JANP, JANPTX and JANPTXV) of MIL-PRF-19500.

	Subgroup	Method	Conditions
	B2	4066	Condition A, 10 Pulses, see 4.5.2 herein.
*	B3	1027	See 4.3.1, 4.5.1, and 4.5.3.
	B5	Not applicable.	
*	B7		See 4.5.4

NOTE: All electrical measurements and tests shall be performed twice, once in each direction.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table J-V of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

* 4.4.3.1 Group C inspection, table J-V of MIL-PRF-19500.

	Subgroup	Method	Conditions
*	C2	1051	-55°C to +125°C
*	C6	1026	See 4.5.1.
	C8		Condition for maximum peak pulse current are as follows: See 4.5.2.b, (20 μs pulse only) 10 pulses; n = 22, c = 0.

NOTE: All electrical measurements and tests shall be performed twice, once in each direction.

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* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table J-VI of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. See table III for delta limits when applicable.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

NOTE: For the purpose of this test, the direction in which the device is first pulsed shall be considered polarity A and the reverse direction polarity B.

* 4.5.1 Group C steady-state operation life test (alternate procedure). When the group B, 340 hour life test is continued on test to 1,000 hours to satisfy the group C life test requirements, the test shall be performed as given in 4.3.1 with the following exceptions: Step 4.3.1.d through 4.3.1.i shall be repeated for an additional 330 hours in each direction for a total of 500 hours in each direction (total = 1,000 hours for group C).

* 4.5.2 Maximum peak pulse current (I_{PP}). The peak pulse currents specified in column 7 of table IV shall be applied while simultaneously maintaining a bias voltage of not less than the applicable voltage specified in column 4 of table IV, in the same polarity as the peak pulse current. The clamping voltage (Vc) shall be as specified in 4.5.3. The peak pulse current shall be applied with a current vs time waveform as follows (1 pulse per minute maximum):

a. Pulse current shall reach 100 percent of I_{PP} at $t \leq 10$ μs and decay to 50 percent of I_{PP} at $t \geq 1$ ms for t_p = 1 ms (see figure 4).

b. Pulse current shall reach 100 percent of I_{PP} at $t \leq 8$ μs and decay to 50 percent of I_{PP} at $t \geq 20$ μs for t_p = 20 μs (see figure 5).

4.5.3 Clamping voltage. The peak pulse clamping voltage shall be measured across the diode in a 1 ms time interval. The response detector shall demonstrate equipment accuracy of ±3 percent. The peak clamping voltage as specified in column 6 of table IV shall be applicable to the 1 ms pulse of 4.5.2.a only.

* 4.5.4 Glass transition temperature. The glass transition temperature test shall be conducted in accordance with MIL-PRF-19500. The use of an alternate test method is permitted provided that the same alternate test method is used to satisfy all instances of glass transition temperature testing requirements within MIL-PRF-19500. Subsequent changes to a different test method shall require a correlation test.

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TABLE I. Group A inspection.

	Inspection 1/	MIL-STD-750		Symbol	Limit 2/	Limit 2/	Unit
		Method	Conditions 2/		Min	Max	
	Subgroup 1						
	Visual and mechanical examination	2071					
	Subgroup 2 3/						
*	Standby current	4016	Condition A or B, DC method, $V_R = V_{WM}$ (column 4 of table IV)	I_D		Column 5	μA dc
	Breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2 percent, I_{BR} = column 3 of table IV	V_{BR}	Column 2	Column 2	V dc
	Subgroup 3 3/						
	Low temperature operation:		T_A = -55°C				
	Minimum breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2 percent, I_{BR} = column 3 of table IV	V_{BR}	Column 10		V dc
	High temperature operation:		T_A = 125°C				
*	Reverse current leakage	4016	Condition A or B, DC method, VR = V_{WM}, column 4 of table IV	I_{D2}		Column 9	mA dc
	Subgroup 4 3/						
*	Clamping voltage maximum		t_p = 1.0 ms (see 4.5.2.a), I_{PP} = column 7 of table IV, (pulsed) (see 4.5.3)	V_C		Column 6	V (pk)
	Subgroup 5, 6 and 7						
	Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ Column references are to table IV.
3/ All electrical testing shall be performed twice, once in each direction.

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* TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

	Inspection	MIL-STD-750		Sampling plan
		Method	Conditions	
	Subgroup 1			45 devices c = 0
	Temperature cycling	1051	500 cycles, condition C, -55°C to +150°C.	
*	Electrical measurements		See table III, steps 1, 2, 3, 4, and 5.	
*	Subgroup 2			45 devices c = 0
	Life test		1,000 hours. See 4.3.1.	
*	Electrical measurements		See table III, steps 1, 2, 3, 4, and 5.	
	Subgroups 4 and 5			
	Not applicable			
	Subgroup 6			
	Peak pulse current		See 4.5.2. I_{PP} shall be characterized by the supplier and this data shall be available to the government. Test shall be performed on each low and high voltage device for each structurally identical grouping. Test to failure.	
*	Electrical measurements		See table III, steps 1, 2, 3, 4, and 5.	
*	Subgroup 10			
	Glass Transition Temperature		See 4.5.4	

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* TABLE III. Groups B and C electrical and delta measurements. 1/ 2/ 3/

	Step	Inspection	MIL-STD-750		Symbol	Limit 4/	Limit 4/	Unit
			Method	Conditions 4/		Min	Max	
*	1.	Standby current	4016	Condition A or B, DC method, $V_R = V_{WM}$ column 4 of table IV	I_D		Column 5	µA dc
	2.	Breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2 percent, I_{BR} = column 3 of table IV	V_{BR}	Column 2	Column 2	V dc
*	3.	Standby current	4016	Condition A or B, DC method; $V_R = V_{WM}$ (column 4 of table IV)	ΔI_D 5/		100 percent of initial reading or 20 percent of column 5 of table IV, whichever is greater.	
	4.	Breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2 percent, I_{BR} = column 3 of table IV	ΔV_{BR} 5/		± 5 percent of initial value	
	5.	Clamping voltage		t_p = 1.0 ms (see 4.5.2.a); I_{PP} = column 7 of table IV	V_C		Column 6	V (pk)

1/ All electrical testing shall be performed twice, once in each direction.
2/ The electrical measurements for table J-IVB of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table III herein, steps 1 and 2.
   b. Subgroup 6 see table III herein, steps 1, 2, 3 and 4.
3/ The electrical measurements for table J-V of MIL-PRF-19500 are as follows:
   a. Subgroup 2 and 3, see table III herein, steps 1 and 2.
   b. Subgroup 9, see table III herein, steps 1 and 2 for all levels.
4/ Column references are to table IV.
5/ Devices which exceed the group A limits for this test shall not be accepted.

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TABLE IV. Characteristics and ratings.

Col 1	Col 2		Col 3	Col 4	Col 5	Col 6	Col 7		Col 8	Col 9	Col 10	Col 11
Type	Breakdown voltage V_{BR} at I_{BR}	Breakdown voltage V_{BR} at I_{BR}	Test current I_{BR}	Working peak reverse voltage V_{WM}	Maximum standby current I_D	Maximum clamping voltage V_C at I_{PP}	Maximum peak pulse current (I_{PP}) t_p = 20 µs t_r = 8 µs	Maximum peak pulse current (I_{PP}) t_p = 1 ms t_r = 10 µs	Maximum temperature coefficient of V_{BR} αV_{BR}	Maximum standby current I_D T_A = +125°C	Minimum breakdown voltage at I_{BR} T_A = -55°C	Breakdown current I_{BR} maximum dc current T_A = +25°C
	Min V dc	Max V dc	mA dc	V (pk)	µA dc	V (pk)	A (pk)	A (pk)	%/°C	µA dc	V dc	mA dc
1N6036A	7.13	7.88	10	6.0	1000	11.3	750	132.0	0.061	2000	6.66	125.0
1N6037A	7.79	8.61	10	7.0	500	12.1	700	124.0	0.065	1000	7.24	115.0
1N6038A	8.65	9.55	10	7.5	200	13.4	630	112.0	0.068	400	8.01	104.0
1N6039A	9.50	10.50	1	8.5	50	14.5	585	103.0	0.073	200	8.75	95.0
1N6040A	10.50	11.60	1	9.0	10	15.6	545	96.0	0.075	200	9.65	86.0
1N6041A	11.40	12.60	1	10.0	5	16.7	510	90.0	0.078	200	10.40	73.0
1N6042A	12.40	13.70	1	11.0	5	18.2	465	82.0	0.081	200	11.30	70.0
1N6043A	14.30	15.80	1	12.0	5	21.2	400	71.0	0.084	200	13.00	63.0
1N6044A	15.20	16.80	1	13.0	5	22.5	375	67.0	0.086	200	13.70	59.0
1N6045A	17.10	18.90	1	15.0	5	25.2	335	59.5	0.088		15.40	53.0
1N6046A	19.00	21.00	1	17.0	5	27.7	305	54.0	0.090	200	17.10	47.0
1N6047A	20.90	23.10	1	18.0	5	30.6	275	49.0	0.092	200	18.80	43.0
1N6048A	22.80	25.20	1	20.0	5	33.2	255	45.0	0.094	200	20.50	39.0
1N6049A	25.70	28.40	1	22.0	5	37.5	225	40.0	0.096	200	23.00	35.0
1N6050A	28.50	31.50	1	25.0	5	41.4	205	36.0	0.097	200	25.50	31.0
1N6051A	31.40	34.70	1	28.0	5	45.7	185	33.0	0.098	200	29.00	28.0
1N6052A	34.20	37.80	1	30.0	5	49.9	170	30.0	0.098	200	30.50	26.0
1N6053A	37.10	41.00	1	33.0	5	53.9	155	28.0	0.100	200	33.10	24.0
1N6054A	40.90	45.20	1	36.0	5	59.3	145	25.3	0.101	200	36.40	22.0
1N6055A	44.70	49.40	1	40.0	5	64.8	130	23.2	0.101	200	39.80	20.0

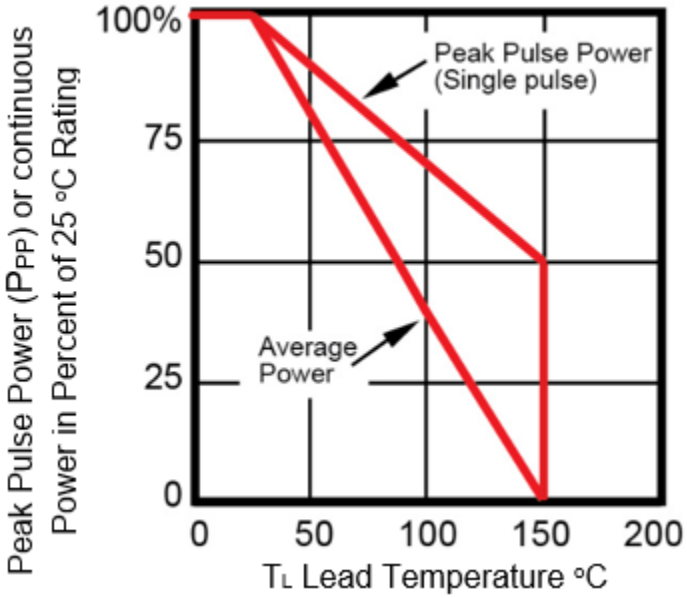
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TABLE IV. Characteristics and ratings - Continued.

Col 1	Col 2		Col 3	Col 4	Col 5	Col 6	Col 7		Col 8	Col 9	Col 10	Col 11
Type	Breakdown voltage V_{BR} at I_{BR}	Breakdown voltage V_{BR} at I_{BR}	Test current I_{BR}	Working peak reverse voltage V_{WM}	Maximum standby current I_D	Maximum clamping voltage V_C at I_{PP}	Maximum peak pulse current (I_{PP}) t_p = 20 µs t_r = 8 µs	Maximum peak pulse current (I_{PP}) t_p = 1 ms t_r = 10 µs	Maximum temperature coefficient of V_{BR} αV_{BR}	Maximum standby current I_D T_A = +125°C	Minimum breakdown voltage at I_{BR} T_A = -55°C	Breakdown current I_{BR} maximum dc current T_A = +25°C
	Min V dc	Max V dc	mA dc	V (pk)	µA dc	V (pk)	A (pk)	A (pk)	%/°C	µA dc	V dc	mA dc
1N6056A	48.50	53.60	1	43.0	5	70.1	120	21.4	0.102	200	43.10	18.0
1N6057A	53.20	58.80	1	47.0	5	77.0	110	19.5	0.103	200	47.30	17.0
1N6058A	58.90	68.20	1	53.0	5	85.0	100	17.7	0.104	200	52.30	15.0
1N6059A	64.60	71.40	1	58.0	5	92.0	90	16.3	0.104	200	57.30	14.0
1N6060A	71.30	78.80	1	64.0	5	103.0	82	14.6	0.105	200	63.20	12.0
1N6061A	77.90	86.10	1	70.0	5	113.0	75	13.3	0.105	200	69.00	11.0
1N6062A	86.50	95.50	1	75.0	5	125.0	68	12.0	0.106	200	76.50	10.0
1N6063A	95.00	105.00	1	82.0	5	137.0	62	11.0	0.106	200	84.10	9.5
1N6064A	105.00	116.00	1	94.0	5	152.0	55	9.9	0.107	200	92.80	8.5
1N6065A	114.00	126.00	1	100.0	5	168.0	50	8.9	0.107	200	100.00	7.5
1N6066A	124.00	137.00	1	110.0	5	182.0	47	8.2	0.107	200	109.00	7.0
1N6067A	143.00	158.00	1	128.0	5	213.0	40	7.0	0.108	200	126.00	6.0
1N6068A	162.00	179.00	1	145.0	5	245.0	36	6.1	0.108	200	143.00	5.5
1N6069A	171.00	189.00	1	150.0	5	261.0	34	5.7	0.108	200	151.00	5.2
1N6070A	181.00	200.00	1	160.0	5	278.0	32	5.4	0.108	200	157.00	5.0
1N6071A	190.00	210.00	1	170.0	5	294.0	30	5.1	0.108	200	169.00	4.7
1N6072A	209.00	231.00	1	185.0	5	328.0	25	4.6	0.108	200	184.00	4.3

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The pulse derating curve of maximum peak pulse power versus temperature has been included for reference purposes only.



* FIGURE 2. Derating curve.

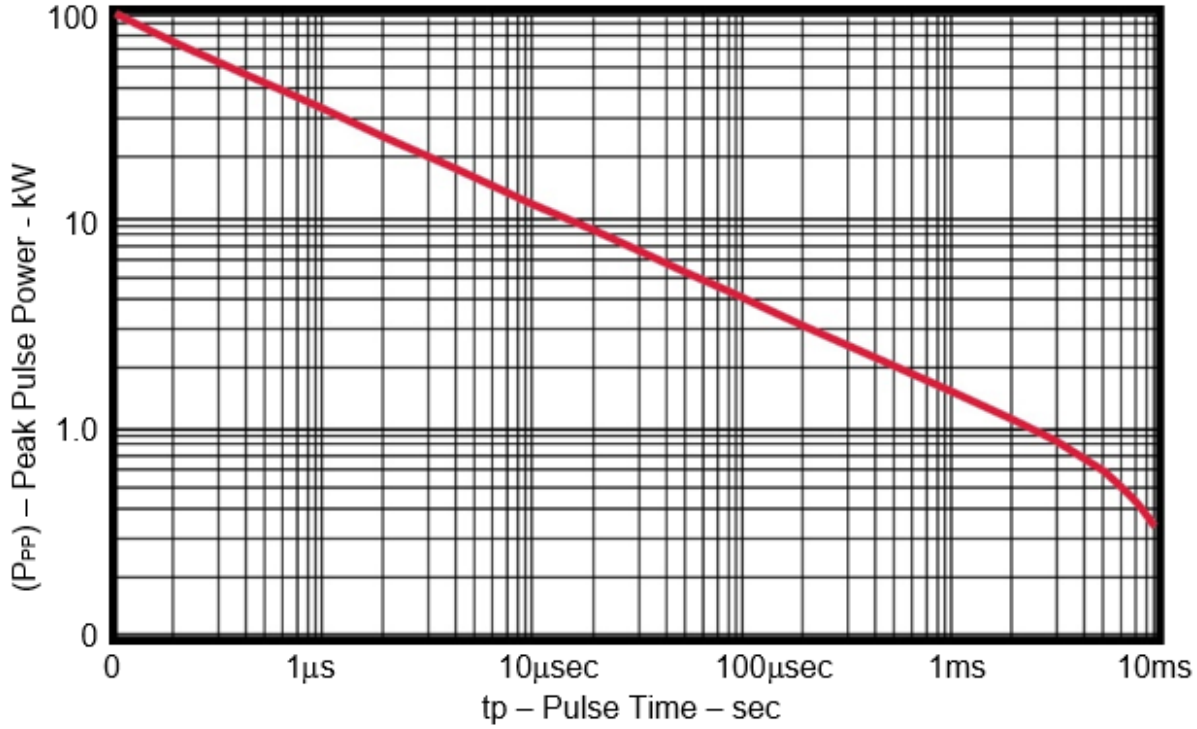
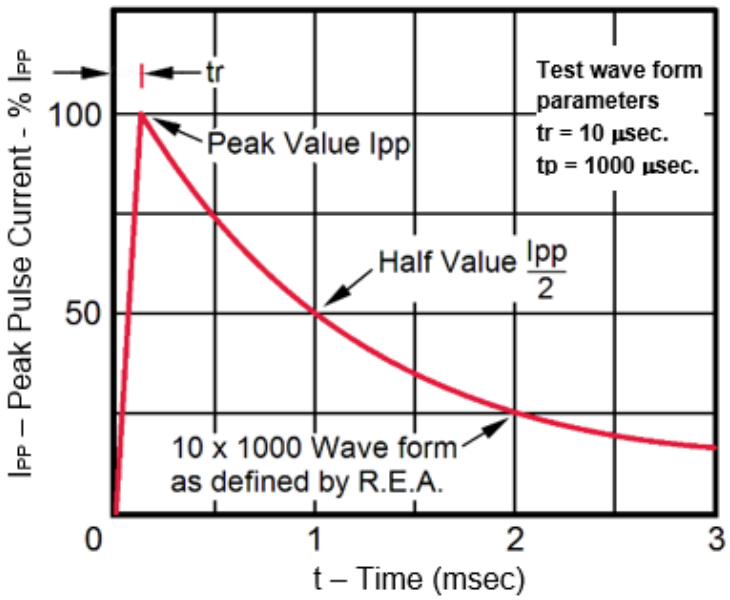


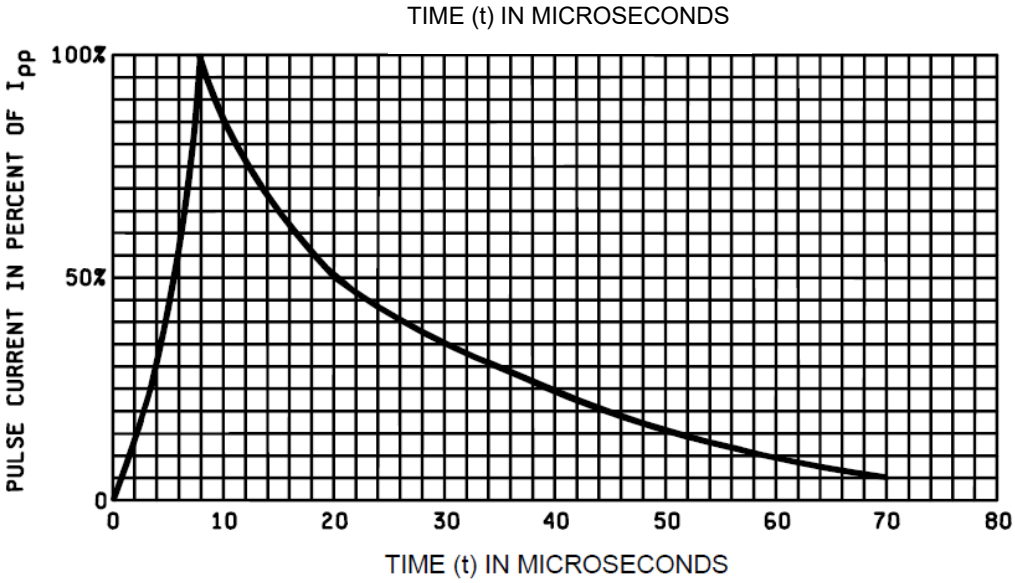
FIGURE 3. Peak pulse power versus pulse time.

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NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of I_{PP}. (Rise time to 100 percent of I_{PP} = 10 µs).

* FIGURE 4. Current impulse waveform (see 4.5.2.a).



NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of I_{PP}. (Rise time to 100 percent of I_{PP} = 8 µs).

* FIGURE 5. Current impulse waveform (see 4.5.2.b).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.) The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

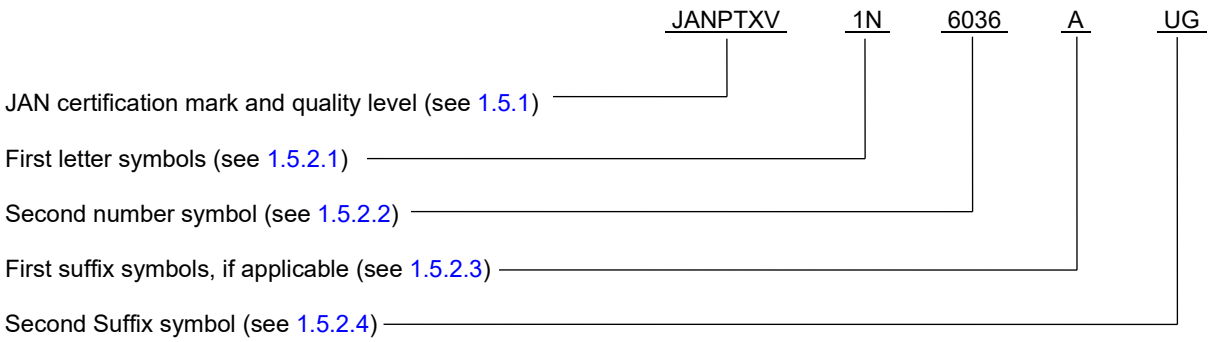
c. Lead finish (see 3.4.2).

* d. The complete PIN, see 1.5 and 6.4.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

* 6.4 PIN Construction Example

* 6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



* 6.5 List of PINs. The following is a list of possible PINs available on this specification sheet. Types 1N6036 through 1N6072.

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JANP1N6036AUG	JANPTX1N6036AUG	JANPTXV1N6036AUG	JANP1N6036AUJ	JANPTX1N6036AUJ	JANPTXV1N6036AUJ
JANP1N6037AUG	JANPTX1N6037AUG	JANPTXV1N6037AUG	JANP1N6037AUJ	JANPTX1N6037AUJ	JANPTXV1N6037AUJ
JANP1N6038AUG	JANPTX1N6038AUG	JANPTXV1N6038AUG	JANP1N6038AUJ	JANPTX1N6038AUJ	JANPTXV1N6038AUJ
JANP1N6039AUG	JANPTX1N6039AUG	JANPTXV1N6039AUG	JANP1N6039AUJ	JANPTX1N6039AUJ	JANPTXV1N6039AUJ
JANP1N6040AUG	JANPTX1N6040AUG	JANPTXV1N6040AUG	JANP1N6040AUJ	JANPTX1N6040AUJ	JANPTXV1N6040AUJ
JANP1N6041AUG	JANPTX1N6041AUG	JANPTXV1N6041AUG	JANP1N6041AUJ	JANPTX1N6041AUJ	JANPTXV1N6041AUJ
JANP1N6042AUG	JANPTX1N6042AUG	JANPTXV1N6042AUG	JANP1N6042AUJ	JANPTX1N6042AUJ	JANPTXV1N6042AUJ
JANP1N6043AUG	JANPTX1N6043AUG	JANPTXV1N6043AUG	JANP1N6043AUJ	JANPTX1N6043AUJ	JANPTXV1N6043AUJ
JANP1N6044AUG	JANPTX1N6044AUG	JANPTXV1N6044AUG	JANP1N6044AUJ	JANPTX1N6044AUJ	JANPTXV1N6044AUJ
JANP1N6045AUG	JANPTX1N6045AUG	JANPTXV1N6045AUG	JANP1N6045AUJ	JANPTX1N6045AUJ	JANPTXV1N6045AUJ
JANP1N6046AUG	JANPTX1N6046AUG	JANPTXV1N6046AUG	JANP1N6046AUJ	JANPTX1N6046AUJ	JANPTXV1N6046AUJ
JANP1N6047AUG	JANPTX1N6047AUG	JANPTXV1N6047AUG	JANP1N6047AUJ	JANPTX1N6047AUJ	JANPTXV1N6047AUJ
JANP1N6048AUG	JANPTX1N6048AUG	JANPTXV1N6048AUG	JANP1N6048AUJ	JANPTX1N6048AUJ	JANPTXV1N6048AUJ
JANP1N6049AUG	JANPTX1N6049AUG	JANPTXV1N6049AUG	JANP1N6049AUJ	JANPTX1N6049AUJ	JANPTXV1N6049AUJ
JANP1N6050AUG	JANPTX1N6050AUG	JANPTXV1N6050AUG	JANP1N6050AUJ	JANPTX1N6050AUJ	JANPTXV1N6050AUJ
JANP1N6051AUG	JANPTX1N6051AUG	JANPTXV1N6051AUG	JANP1N6051AUJ	JANPTX1N6051AUJ	JANPTXV1N6051AUJ
JANP1N6052AUG	JANPTX1N6052AUG	JANPTXV1N6052AUG	JANP1N6052AUJ	JANPTX1N6052AUJ	JANPTXV1N6052AUJ
JANP1N6053AUG	JANPTX1N6053AUG	JANPTXV1N6053AUG	JANP1N6053AUJ	JANPTX1N6053AUJ	JANPTXV1N6053AUJ
JANP1N6054AUG	JANPTX1N6054AUG	JANPTXV1N6054AUG	JANP1N6054AUJ	JANPTX1N6054AUJ	JANPTXV1N6054AUJ
JANP1N6055AUG	JANPTX1N6055AUG	JANPTXV1N6055AUG	JANP1N6055AUJ	JANPTX1N6055AUJ	JANPTXV1N6055AUJ
JANP1N6056AUG	JANPTX1N6056AUG	JANPTXV1N6056AUG	JANP1N6056AUJ	JANPTX1N6056AUJ	JANPTXV1N6056AUJ
JANP1N6057AUG	JANPTX1N6057AUG	JANPTXV1N6057AUG	JANP1N6057AUJ	JANPTX1N6057AUJ	JANPTXV1N6057AUJ
JANP1N6058AUG	JANPTX1N6058AUG	JANPTXV1N6058AUG	JANP1N6058AUJ	JANPTX1N6058AUJ	JANPTXV1N6058AUJ
JANP1N6059AUG	JANPTX1N6059AUG	JANPTXV1N6059AUG	JANP1N6059AUJ	JANPTX1N6059AUJ	JANPTXV1N6059AUJ
JANP1N6060AUG	JANPTX1N6060AUG	JANPTXV1N6060AUG	JANP1N6060AUJ	JANPTX1N6060AUJ	JANPTXV1N6060AUJ
JANP1N6061AUG	JANPTX1N6061AUG	JANPTXV1N6061AUG	JANP1N6061AUJ	JANPTX1N6061AUJ	JANPTXV1N6061AUJ
JANP1N6062AUG	JANPTX1N6062AUG	JANPTXV1N6062AUG	JANP1N6062AUJ	JANPTX1N6062AUJ	JANPTXV1N6062AUJ
JANP1N6063AUG	JANPTX1N6063AUG	JANPTXV1N6063AUG	JANP1N6063AUJ	JANPTX1N6063AUJ	JANPTXV1N6063AUJ
JANP1N6064AUG	JANPTX1N6064AUG	JANPTXV1N6064AUG	JANP1N6064AUJ	JANPTX1N6064AUJ	JANPTXV1N6064AUJ
JANP1N6065AUG	JANPTX1N6065AUG	JANPTXV1N6065AUG	JANP1N6065AUJ	JANPTX1N6065AUJ	JANPTXV1N6065AUJ
JANP1N6066AUG	JANPTX1N6066AUG	JANPTXV1N6066AUG	JANP1N6066AUJ	JANPTX1N6066AUJ	JANPTXV1N6066AUJ
JANP1N6067AUG	JANPTX1N6067AUG	JANPTXV1N6067AUG	JANP1N6067AUJ	JANPTX1N6067AUJ	JANPTXV1N6067AUJ
JANP1N6068AUG	JANPTX1N6068AUG	JANPTXV1N6068AUG	JANP1N6068AUJ	JANPTX1N6068AUJ	JANPTXV1N6068AUJ
JANP1N6069AUG	JANPTX1N6069AUG	JANPTXV1N6069AUG	JANP1N6069AUJ	JANPTX1N6069AUJ	JANPTXV1N6069AUJ
JANP1N6070AUG	JANPTX1N6070AUG	JANPTXV1N6070AUG	JANP1N6070AUJ	JANPTX1N6070AUJ	JANPTXV1N6070AUJ
JANP1N6071AUG	JANPTX1N6071AUG	JANPTXV1N6071AUG	JANP1N6071AUJ	JANPTX1N6071AUJ	JANPTXV1N6071AUJ
JANP1N6072AUG	JANPTX1N6072AUG	JANPTXV1N6072AUG	JANP1N6072AUJ	JANPTX1N6072AUJ	JANPTXV1N6072AUJ

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* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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